CT 01 ICT 4101 Data Mining Marks: 20

1. Mention few sources of data impurities. Why is missing data not good? 5
2. What are the different data attribute types? Write some examples. 5
3. Quartiles important in data discovery. How, and why? 5
4. What qualities define data quality? Why do they matter? 5

CT 02 **ICT 4101** **Data Mining** **Marks: 20**

1. What do you understand by "mining frequent itemset mining"? 5
2. Briefly explain the concept of market basket modeling and its applications. 5
3. How does the Apriori algorithm handle candidate generation and pruning? 5
4. Define support and confidence. 5

CT 03 **ICT 4101** **Data Mining** **Marks: 20**

1. Describe few advantages of unsupervised learning over supervised learning. 5
2. Briefly write about information and expected information. 5
3. Why do you need a hyperplane in the SVM? 5
4. Do you think that multiple linear regression is better than linear regression? 5

CT 04 **ICT 4101** **Data Mining** **Marks: 20**

1. How is clustering different from the classification? 5
2. How many ways can data be clustered? 5
3. How many clustering models are there? 5
4. What are the challenges in high dimension data clustering? 5

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Lab Quiz **ICT 4102** **Data Mining** **Marks: 20**

1. Distance measurements: (i) Euclidian (ii) Jaccard distances. 5

2. 5

	Blues Traveler	Norah Jones	Phoenix	The Strokes	Weird Al
Clara	4.75	4.5	5	4.25	4
Robert	4	3	5	2	1

Find cosine similarity for the above data.

3.

	Angelica	Bill
Blues Traveler	3.5	2
Broken Bells	2	3.5
Deadmau5	-	4
Norah Jones	4.5	-
Phoenix	5	2
Slightly Stoopid	1.5	3.5
The Strokes	2.5	-
Vampire Weekend	2	3

Compute Euclidean distance for the above data. 5

4. How Minkowski distance is related to Euclidean and Manhattan distances? 5

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4th Year 1st Semester Final Examination 2025

Course ID: ICT 4102 Course Title: Data Mining Lab

Answer all questions | May use GenAI

1. a) Use an example to show why the *k*-means algorithm may not find the global optimum, that is, optimizing the within-cluster variation. 10

   b) Does the algorithm reach a global optimum? Verify with python program and show the result.

2. a) Traditional clustering methods are rigid in that they require each object to belong exclusively to only one cluster. Explain why this is a special case of fuzzy clustering. You may use *k*-means as an example. 10

   b) To interpret the fuzzy result, use python programing.

OR

2. a) It is difficult to assess classification accuracy when individual data objects may belong to more than one class at a time. In such cases, comment on what criteria you would use to compare different classifiers modeled after the same data. 10

   b) Is the Hamming loss better? Why? Verify with python program and show the result.

3. a) The following table summarizes supermarket transaction data, where *hot chili* refers to the transactions containing hot chilis, *not hot chili* refers to the transactions that do not contain hot chilis, *spaghetti* refers to the transactions containing spaghetti, and *not spaghetti* refers to the transactions that do not contain spaghetti. 20

	Hot chili	Not hot chili	Count
Spaghetti	2000	500	2500
Not spaghetti	1000	1500	2500
Sum	3000	2000	5000

   i) Suppose that the association rule "hot chili ⇒ spaghetti" is mined. Given a minimum support threshold of 25% and a minimum confidence threshold of 50%, is this association rule strong?

   ii) Based on the given data, is the purchase of hot chilis independent of the purchase of hamburgers? If not, what kind of correlation relationship exists between the two?

   iii) Compare the use of all confidence, max confidence, and cosine measures with lift and correlation on the given data.

   b) Find the frequent dataset.

   c) Write a python program and show all results.

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